Waseem Raza, PhD

Machine Learning Engineer | Financial AI/ML & Data Science

Financial ML | Forecasting | Risk Analytics | Data Engineering

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LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Professional Profile

Machine Learning Engineer and Data Scientist with a PhD in Electrical and Computer Engineering, production ML experience, and finance-focused portfolio work in credit-risk modeling, loan-default prediction, time-series forecasting, and reinforcement-learning applications. Skilled in Python, SQL, scikit-learn, XGBoost, TensorFlow, PyTorch, Pandas, NumPy, statistical modeling, cross-validation, feature engineering, Azure Databricks, Apache Spark, ETL/ELT, predictive analytics, and model evaluation. Developing deeper specialization in financial forecasting, risk analytics, portfolio optimization, and data-driven decision intelligence.

Technical Competencies

Financial Machine Learning: Credit-risk modeling, loan-default prediction, supervised learning, predictive modeling, reinforcement-learning concepts, portfolio and trading applications

Time Series & Statistics: Forecasting, statistical modeling, feature engineering, regression, classification, model diagnostics, validation, and uncertainty-aware analysis

Data Science & Analytics: Python, SQL, Pandas, NumPy, SciPy, Statsmodels, exploratory analysis, visualization, and data-quality assessment

Machine Learning: scikit-learn, XGBoost, TensorFlow, PyTorch, cross-validation, hyperparameter tuning, anomaly detection, and model evaluation

Data Engineering & Cloud: ETL/ELT, Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Synapse Analytics, BigQuery, and AI-ready datasets

MLOps & Engineering: Git, Docker, CI/CD, APIs, testing, reproducible workflows, model deployment, and production-oriented engineering

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer **02/2025 - Present**

AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Lead end-to-end development, validation, and operational integration of machine-learning diagnostics, including data preparation, classification, model evaluation, and error analysis.
- Design domain-adaptive neural models and reusable data-processing pipelines that improve robustness across heterogeneous operational datasets and distribution shifts.
- Build cross-domain validation, data-quality, and failure-mode assessment workflows to support reliable predictive analytics and decision-support systems.
- Translate complex operational requirements into measurable ML objectives, validated datasets, and production-oriented analytical workflows.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**

Impac Exploration Services *Weatherford, OK*

- Developed scalable analytical and ETL pipelines using Python, Azure Databricks, Azure Functions, and Synapse Analytics for large structured datasets.
- Automated ingestion, cleansing, transformation, and validation workflows, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Translated domain requirements into AI-ready datasets and reproducible analytical workflows for data-driven decision support.

Advanced-Degrees Intern **06/2024 - 08/2024**

AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.

- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Graduate Research Assistant **09/2019 - 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Projects & Research

Projects & Applied Research

Financial ML and Forecasting *Portfolio Project | 2024-2025*
Developed finance-oriented machine-learning workflows for credit-risk and loan-default modeling together with time-series forecasting experiments. Applied data preparation, feature engineering, model comparison, cross-validation, and predictive-performance analysis to structured financial datasets.
Tools & Methods: Python, scikit-learn, Classification, Credit Risk, Loan Default Modeling, Time-Series Forecasting, Feature Engineering, Cross-Validation, Model Evaluation

AI-Ready Lakehouse Pipeline *Portfolio Project | 2024*
Designed an AI-ready lakehouse pipeline that prepares structured and semi-structured data for analytics and downstream machine-learning workloads. Implemented repeatable ingestion, cleansing, transformation, validation, and scalable processing patterns suitable for reproducible cloud data workflows.
Tools & Methods: Azure Databricks, Apache Spark, PySpark, ETL/ELT, Data Ingestion, Schema Transformation, Data Validation, Feature Preparation, Cloud Data Engineering

System Design Case-Study Portfolio *Guided Project | 2026*
Developed architecture case studies that translate ambiguous product requirements and traffic assumptions into service boundaries, APIs, data models, storage choices, caching strategies, and reliability targets. Documented scalability, consistency, capacity, and failure-mode tradeoffs to support production-oriented system design decisions.
Tools & Methods: System Design, REST APIs, Distributed Systems, Data Modeling, Caching, Scalability, Reliability, Capacity Planning, Architecture Tradeoffs

Agentic AI and RAG Applications *Portfolio Project | 2026*
Built grounded GenAI workflows that combine document retrieval, vector search, stateful orchestration, tool-using agents, memory, evaluation, and multi-agent handoffs. Emphasized controllable execution, traceable context flow, and reliable responses for multi-step tasks rather than single unconstrained model calls.
Tools & Methods: Python, LangGraph, LangChain, CrewAI, AutoGen, RAG, Vector Databases, Embeddings, Tool Calling, Agent Memory, Evaluation

OCR & Document Intelligence Extraction Pipeline *Guided Project | 2025*
Implemented a document-intelligence pipeline that converts reports, invoices, and contract-style documents into structured outputs through OCR, layout/table parsing, entity and field extraction, normalization, and validation. Designed the workflow for downstream analytics and data integration rather than isolated text extraction.
Tools & Methods: OCR, Document Intelligence, LLMs, Named Entity Recognition, Table Extraction, Structured JSON, Normalization, Validation, Data Integration

Amazon Bedrock Generative AI & Agent Workflows *Certification Project | 2026*
Implemented foundation-model prompting, knowledge-base retrieval, RAG integration, model-customization exercises, workflow automation, and intelligent-agent patterns using Amazon Bedrock. Explored production-oriented cloud GenAI patterns with managed model access, retrieval-backed grounding, and reusable application workflows.

Tools & Methods: Amazon Bedrock, Foundation Models, RAG, Knowledge Bases, LangChain, AI Agents, Prompt Engineering, AWS GenAI Services, Workflow Automation

Publications

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *2025*
Sensors 2025

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *2024*
IEEE PIMRC

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *2024*
IEEE SmartNets

Certifications & Professional Development

1. Machine Learning — Stanford Online / Coursera | Completed | March 29, 2020

Focus: Covers foundational machine-learning methods for supervised and unsupervised problems, including regression, classification, model evaluation, regularization, clustering, and practical model-selection considerations. Emphasizes translating data into predictive models while controlling generalization error.

Tools & Skills: supervised learning, regression, classification, regularization, clustering, model evaluation, bias-variance tradeoffs, machine-learning workflows

2. Tesla Stock Price Prediction using Facebook Prophet — Coursera | Completed | March 16, 2026

Focus: Applies time-series forecasting to stock-price data using Facebook Prophet, covering data preparation, trend and seasonality modeling, forecast generation, and visualization. Demonstrates an end-to-end predictive workflow for financial time-series analysis.

Tools & Skills: time-series forecasting, Facebook Prophet, financial data, trend analysis, seasonality, forecast visualization, predictive analytics

3. Introduction to Data Analytics in Google Cloud — Coursera | Completed | July 24, 2024

Focus: Introduces cloud-native data analytics in Google Cloud, covering the analytics lifecycle, data sources, storage and processing concepts, and how cloud services support scalable analysis. Emphasizes organizing data work to produce reliable analytical insight.

Tools & Skills: Google Cloud, cloud analytics, data lifecycle, data-ingestion concepts, storage and processing, analytical workflows, data-driven decision support

4. Python for Data Science, AI & Development — IBM / Coursera | Completed | July 12, 2026 | Certificate

Focus: Builds practical Python skills for data science and AI development, including core programming, data structures, data manipulation, and application-oriented analytical workflows. Emphasizes using Python as a reusable foundation for data processing, modeling, and AI applications.

Tools & Skills: Python, data structures, data manipulation, Pandas concepts, NumPy concepts, data-analysis workflows, AI application development

5. Exploratory vs Confirmatory data analysis using Python — Coursera | Completed | August 21, 2024

Focus: Distinguishes exploratory data analysis from confirmatory analysis and applies Python-based statistical workflows to investigate patterns and test hypotheses. Emphasizes selecting analytical methods that match the purpose and stage of a data-science investigation.

Tools & Skills: exploratory data analysis, confirmatory analysis, Python, statistical reasoning, data visualization, hypothesis-testing concepts, analytical workflow design

6. AI Fundamentals in Financial Services — Coursera

Focus: Current study focuses on AI applications in financial services, including how data-driven systems can support customer, operational, risk, and analytical workflows. Emphasizes practical opportunities, limitations, and responsible adoption of AI in financial organizations.

Tools & Skills: financial-services AI, AI use-case analysis, risk and operations concepts, data-driven decision support, responsible AI, financial analytics

7. Fundamentals of Machine Learning in Finance — Coursera

Focus: Current study focuses on machine-learning foundations for financial data, connecting supervised-learning concepts, feature design, model validation, and predictive tasks with finance-oriented examples. Emphasizes robust modeling choices for noisy and decision-sensitive datasets.

Tools & Skills: machine learning in finance, supervised learning, feature engineering, model validation, predictive analytics, financial-data modeling